

Department of Electrical and Computer Engineering, North South University

CSE 299: Junior Design Course

Fall 2026 | Section: 13 & 14 | Faculty: SMSL

Start With a Real Problem

Before choosing a track, think of a real problem you genuinely want to work on. It can be something that you, your family, your campus, or your community actually face, preferably one that is specific to the local context. Once you have a list of ideas, consider the tracks mentioned below and refine your ideas accordingly. If appropriate, you should also talk to the people affected by the problem and observe it firsthand.

Project Tracks

Track 1: Intelligent Systems for Local Problems. Build an AI-based system that solves a real problem in your local context. Your system should be able to remember information, make decisions in multiple steps, and use different tools or sources of information. You may use free AI APIs where appropriate. Test your system on realistic examples and report how long it takes to respond and how much it costs to run.

Track 2: Simulation and Systems Study. Choose a real-world system, build a computer model of it, and collect real data to check whether your model behaves realistically. Then use the model to explore “what if” questions. An LLM or a machine learning model should make decisions inside the simulation, and you have to compare its decisions with simpler, traditional approaches. Run the simulation multiple times, report how consistent the results are, and end with a recommendation supported by your results.

Track 3: ML Model and Dataset Development. Collect and label a new dataset based on a real local problem, with clear labeling guidelines and a short description of the data and its limitations. Train and compare at least three approaches: a traditional ML method, a fine-tuned pretrained model, and a zero-shot LLM (API). Analyze where the models make mistakes and test how well they perform on new or changed data.

Across all tracks: work with data you collect or build for a user you actually serve; explain in your proposal what makes the project technically challenging; and honestly report where your system may fail.

Evaluation

Assessment	%
Proposal Presentation	10
Regular Assessment	20
Mid-Project Report	15
Live Demo	25
Final Presentation	15
Final Report	15

Timeline

Week	Activity
Week 1	Introduction to the course and tracks
Week 2	Bring ideas: real problems you have identified
Week 3	Proposal presentation
Weeks 4–6	Progress updates
Week 7	Mid-Project Report
Weeks 8–10	Progress updates
Weeks 11–12	Final presentation and live demo; final report submission